

National Land Acquisition & Management System

SIH 2026 Hackathon Team Study Handbook

1. Team Members & Technical Part Assignments

This handbook outlines each team member's role and technical assignments for the NLAMS prototype. Study your corresponding section below to understand your part of the project code and database design:

Member 1: React UI & Dashboard Developer

Responsible for the visual layout, user dashboard, styling, and localization switchers. Key assignments include: React state hooks (useState, useEffect) for dynamic API binding; implementing the saffron, navy, and emerald government light-theme visual language; building the global Hindi/English translation maps (Translation.js) for all labels, timelines, forms, and buttons; and rendering the predictive graphs and compensation progress bars using Recharts SVG charts.

Member 2: Backend Core & Database Engineer

Responsible for the server endpoints, relational database design, data persistence, and seeding. Key assignments include: setting up FastAPI async routing and controllers; designing the sqlalchemy ORM tables with project, parcel, and beneficiary schemas; implementing parameterised query guards to eliminate SQL injection risks; enforcing secure write-access checks using API keys; and executing the sqlite seeding script to populate the database with a realistic test set of 2,501 records.

Member 3 (Shivam): Workflow State Machine & ML Specialist

Responsible for business validation logic, the LARR workflow state machine, and delay predictions. Key assignments include: coding the LARR sequence validation (Submitted -> GIS check -> Sec 11 legal notice -> Award -> Handover); pre-processing raw SQLite project data with Pandas; training the Scikit-Learn Classifier and Regressor models; saving weights to land_delay_model.joblib; and deploying the GET /projects/{id}/predict REST API endpoint to output live machine learning delay risk predictions.

Member 4: GIS Engine & Mobile Field UI Developer

Responsible for geographical mapping overlays, GeoJSON geometry, and mobile responsiveness. Key assignments include: integrating Leaflet maps inside React Components; parsing and rendering parcel GeoJSON boundaries as styled interactive polygons; color-coding parcels dynamically by status (Green = Acquired, Orange = Pending, Red = Disputed); and implementing HTML5 Geolocation sensors to allow surveyors to capture coordinates on-site with touch-friendly layout reflows.

Member 5: Security, RBAC & Audit Logger

Responsible for cybersecurity hardening, payload sanitization, and immutable logging. Key assignments include: implementing password hashing algorithms (bcrypt via passlib); coding global FastAPI middleware to strip HTML/Script tags from incoming payloads to prevent stored XSS attacks; configuring strict CORS origin lists to limit access to approved local ports and Cloudflare tunnels; and checking wallet address signatures during MetaMask DBT escrow fund disbursements.

Member 6: QA Testing, MIS Reports & Presentation Lead

Responsible for testing, compiling summaries, generating files, and structure of the deck. Key assignments include: writing the frontend MIS Exporter button to parse project arrays and compile them into downloadable CSV spreadsheets; running edge-case API tests via Swagger UI; validating successful build operations across the Vite compiler; and structuring the 3-minute pitch deck narrative explaining the system's operational pillars and real-world impact to the judges.

2. Website Architecture & Layout Features

- **Tricolor Home & Directives:** Official national entry portal showcasing the LARR Act 2013 timeline and animated highlights of operational pillars.
- **Bilingual Language Switcher:** A globe selector in the navbar translating charts, forms, and official Gazette forms dynamically to Hindi.
- **Executive Dashboard & GIS:** Renders land parcels as Leaflet map overlay polygons alongside Recharts predictive dashboards.
- **Automated Workflows:** A pipeline allowing Central, State, and District roles to sign off on stages, instantly generating printable official Gazette templates.
- **Web3 Trust Portal:** Simulates smart contract escrow releases with Metamask wallet connectivity, a digital deed SHA-256 validation scanner, and a scrolling block ledger.
- **Mobile Field Survey Node:** Allows field surveyors to capture coordinate coordinates using GPS triangulation and report soil classification, structures, and site images.
- **SMS Alert Simulator:** Fires dynamic, floating mobile network text notification alerts during proposal submissions, boundary checks, and DBT escrow releases.
- **AI Delay Forecasting:** The Scikit-Learn model endpoint checking administrative bottlenecks.

3. Technical Stack Breakdown

- **Frontend Framework:** React.js built on Vite (Vite 8, React 19) for immediate loading times and HMR (Hot Module Replacement).
- **Styling Stack:** Tailwind CSS v4 for fully custom responsive components, clean government-themed color palettes, and layouts.
- **GIS Map Rendering:** React Leaflet & Leaflet.js utilizing OpenStreetMap tile layers to draw and render coordinate boundaries (polygons).
- **MIS Visual Charts:** Recharts package using SVG elements for responsive financial and progress bar graphs.
- **Web3 Simulation:** Context-based mock ledger engine simulating SHA-256 hashes, transaction block creation, and crypto signatures.

4. Pitch Defense Strategy for Team Members

- **React UI (M1):** UI-based role selectors are for operator convenience and workflow simulation, while the actual API routes are secured backend-side through cryptographic checks and middleware.
- **Backend (M2):** Database transactions use sqlmodel parameterized queries to prevent SQL injections, and strict XSS sanitization checks are enforced on all text columns.

- **Workflow & ML (Shivam):** The Random Forest/Decision Tree model was trained using the 2,501 SQLite records on features (State, Area, Budget) and optimizes Recall to ensure bottlenecks are flagged early.
- **GIS (M4):** Physical boundaries submitted by field surveyors are cross-checked against registered cadastral database shapes (BhuNaksha) to reject overlaps.
- **Security (M5):** Database audit logs use append-only constraints with zero UPDATE or DELETE permissions, guaranteeing a cryptographically tamper-proof paper trail.